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MLP-Initialized Reinforcement Learning for Exit Timing in SPX Short-Straddle Portfolios

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PROBLEM When should an existing SPX short straddle be closed?

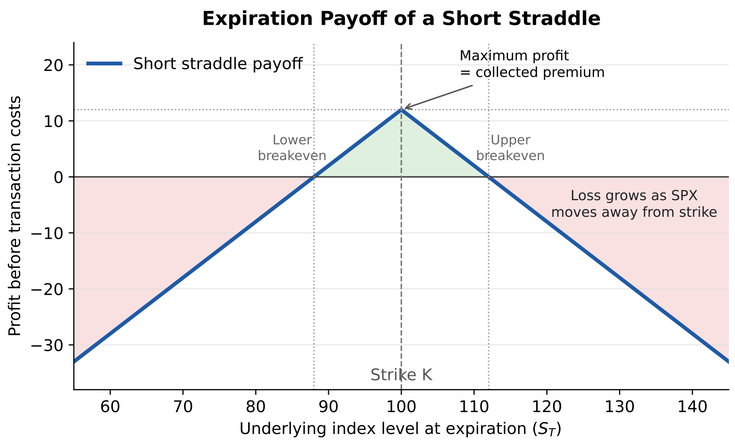
METHOD Learn historical optimal exits with an MLP, then initialize reward-based RL fine-tuning.

RESULT Supervised initialization is strong; the incremental RL gain is positive but inconclusive.

01 Why does exit timing matter?

Collect premium, retain tail risk.

A short straddle sells a call and put with the same strike and expiry. Large SPX moves can overwhelm the premium collected.

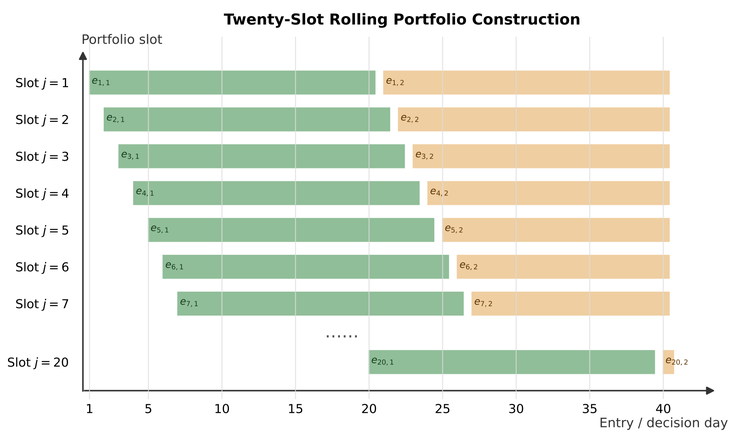


Illustrative expiration payoff before transaction costs; not an estimated empirical response.

The decision is hold or exit. Entry is mechanical and common across policies. The learned policy controls when an already-open position is closed.

02 How is capital reused?

Twenty sleeves; one position per sleeve.



Schematic, not observed trade durations. Exited or expired sleeves can be reused; actual exits vary by episode.

$3m initial capital; stress-budget position sizing and common margin rules across policies.

03 What is the test protocol?

Cboe SPX, 2016–2026. 30-minute quotes, implied volatility, Greeks and underlying index levels.

Pre-test development: 2016–2023.

MLP: internal 80/20 state-sample train/validation split; final RL fine-tuning uses 2016–2023.

Reward coefficients inherited from a preliminary search: train 2016–2021; validate 2022–2023 (500 episodes).

Locked test: Jan 2, 2024–Jun 5, 2026; 7,868 real-grid timestamps.

- Mechanical near-ATM entry at about 20 trading days to maturity; 26-bar minimum hold (about 2 trading days).
- Sell at bid, close at ask; $1.50 per option leg/contract/event; cash settlement.

Current-state features. Maturity, VRP, spreads, volatility changes, moneyness, PnL, and quote/synthetic-bar flags.

04 How are exits learned?

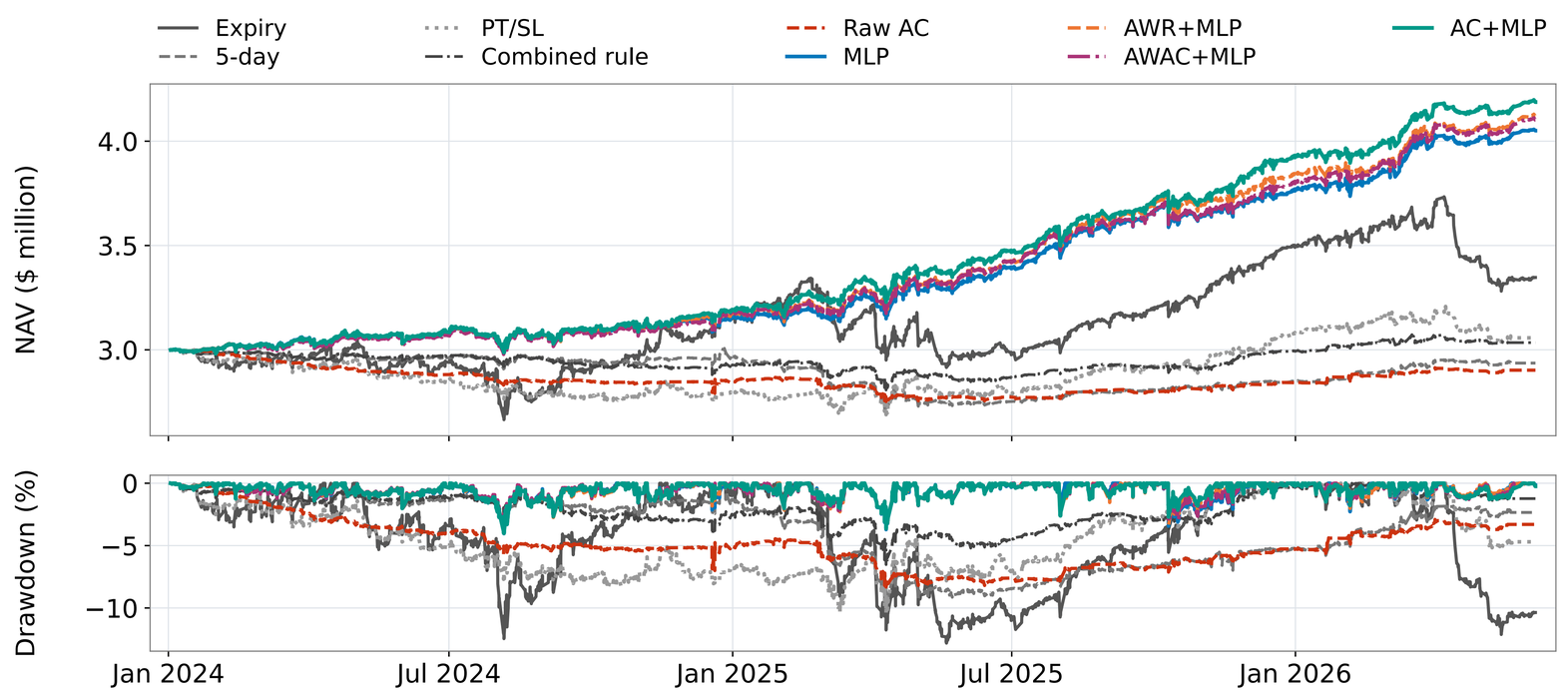
Historical optimal exits
$\tau_i^\star = \arg\max_{26 \le t \le T_i} \Pi_{i,t}^{\text{exit}}$
→ **Supervised MLP**
17 current-state features
Hold/exit
→ **MLP-initialized RL**
AC / AWR / AWAC-style
fine-tuning

Supervised stage. Two ReLU layers; weighted cross-entropy on $y_{i,t} = \mathbf{1}\{t \ge \tau_i^\star\}$.

RL stage. PnL/mark-to-market rewards with VRP, liquidity-cost, CVaR and late-holding terms.

Future paths construct offline labels only. Test policies use contemporaneous information; scaling uses training moments only.

05 How do the paths compare?



Historical representative runs, not seed averages; common stress-budget rules. Cross-seed summaries: Section 06.

06 What do the numbers show?

Policy	Return (%)	Sharpe	MaxDD (%)
Hold to expiry	11.56	0.441	12.82
Fixed 5-day	-2.12	-0.150	9.78
PT / SL	1.91	0.144	10.45
Combined rule	1.15	0.133	6.52
Raw AC	0.18 ± 4.35	-0.102 ± 0.339	8.70 ± 2.67
MLP	35.57 ± 1.41	1.603 ± 0.068	4.34 ± 0.31
AWR+MLP	35.23 ± 1.93	1.648 ± 0.045	3.95 ± 0.11
AWAC+MLP	35.19 ± 1.76	1.655 ± 0.053	3.88 ± 0.05
AC+MLP	$\mathbf{37.54 \pm 0.92}$	$\mathbf{1.702 \pm 0.028}$	$\mathbf{3.96 \pm 0.16}$

Returns are cumulative over the 2.42-year OOS window; MaxDD is loss magnitude. PT/SL: 25%/50%. Learned policies: mean ± sample SD; rules: single paths. No risk-free adjustment to Sharpe. Paper Tables 1–2.

07 What drives the improvement?

Earlier exits. Mean holding time: 18.2 days for passive versus about 10.5 for MLP/AC+MLP (trading days).

Oracle ceiling. Perfect foresight, non-tradable: 373.02% cumulative return; Sharpe 6.305.

Initialization. All seven MLP seeds beat passive; raw AC averages 0.18%.

Seeds: MLP/AC+MLP 7 each; raw AC/AWR/AWAC 5 each. AC+MLP SD excludes supervised-stage variance (fixed MLP).

08 Does RL add value?

Positive, statistically inconclusive. AC+MLP adds 1.97 percentage points in mean return and 0.099 in Sharpe over MLP.

Matched-episode evidence is mixed. 583 matched entered episodes: about $74 additional PnL; paired t-test $p = 0.321$; Wilcoxon $p = 0.027$. PnL averages use available runs with actual entry. Overlapping episodes limit inference.

Equal-notional sizing: the increment falls to 0.45 pp ($p = 0.645$). Dynamic sizing amplifies the gain.

Reward sensitivity. Results are broadly similar without CVaR or VRP, or with direct rewards.

09 Limits and further resources

No sustained crisis-scale shock is tested. Observed drawdowns do not bound tail risk. Execution omits latency, depth, partial fills and market impact.

Audit scope: synthetic-fill checks cover MLP and one representative AC+MLP path only.

Full paper

Code

arXiv: 2610.13126

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Wang, J., Ślepaczuk, R., & Sakowski, P. (2026). *MLP-Initialized Reinforcement Learning for Exit Timing in SPX Short-Straddle Portfolios*. arXiv:2610.13126. https://doi.org/10.48550/arXiv.2610.13126